

NAME

subsets -- Description of CCD subsets

DESCRIPTION

The **ccdred** package groups observation into subsets. The image header parameter used to identify the subsets is defined in the instrument translation file (see help for **instruments**). For example to select subsets by the header parameter "filters" the instrument translation file would contain the line:

```
subset filters
```

Observations are generally grouped into subsets based on a instrument configuration such as a filter, aperture mask, grating setting, etc. This allows combining images from several different subsets automatically and applying the appropriate flat field image when processing the observations. For example if the subsets are by filter then **flatcombine** will search through all the images, find the flat field images (based on the CCD type parameter), and combine the flat field images from each filter separately. Then when processing the images the flat field with the same filter as the observation is used.

Each subset is assigned a short identifier. This is listed when using **ccdlist** and is appended to a root name when combining images. Because the subset parameter in the image header may be any string there must be a mapping applied to generate unique identifiers. This mapping is defined in the file given by the package parameter *ccdred.ssfile*. The file consists of lines with two fields (except that comment lines may be included as a line by itself or following the second field):

```
'subset string' subset_id
```

where the subset string is the image header string and the subset_id is the identifier. A field must be quoted if it contains blanks. The user may create this file but generally it is created by the tasks. The tasks use the first word of the subset string as the default identifier and a number is appended if the first word is not unique. The following steps define the subset identifier:

- (1) Search the subset file, if present, for a matching subset string and use the defined subset identifier.
- (2) If there is no matching subset string use the first word of the image header subset string and, if it is not unique, add successive integers until it is unique.
- (3) If the identifier is not in the subset file create the file and add an entry if necessary.

EXAMPLES

1. The subset file is "subsets" (the default). The subset parameter is translated to "f1pos" in the image header (the old NOAO CCD parameter) which is an integer filter position. After running a "ccdlist *.imh" to cause all filters to be checked, the subset file contains:

```
'2'      2
'5'      5
'3'      3
```

The order reflects the order in which the filters were encountered. Suppose the user wants to have more descriptive names then the subset file can be created or edited to the form:

```
# Sample translation file.
'2'      U
'3'      B
'4'      V
```

(This is only an example and does not mean these are standard filters.)

2. As another example suppose the image header parameter is "filter" and contains more descriptive strings. The subset file might become:

```
'GG 385 Filter' GG
'GG 495 Filter' GG1
'RG 610 Filter' RG
'H-ALPHA'      H_ALPHA
```

In this case use of the first word was not very good but it is better if the filters are encoded with the thought that the first word will be used by **ccdred**; it should be short and unique.

SEE ALSO

instruments